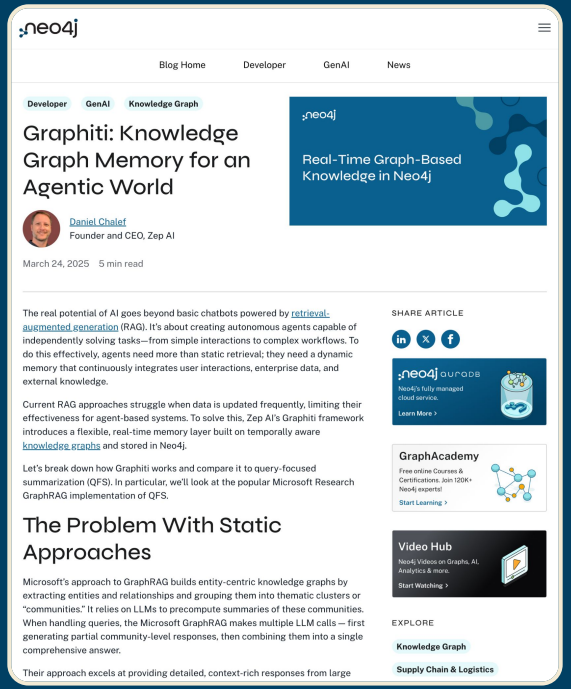
neo4j

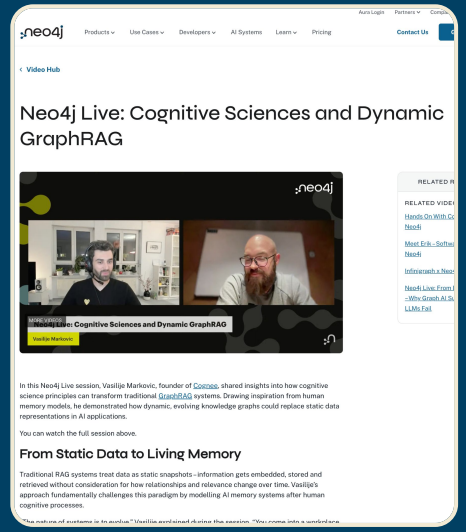
Going Meta

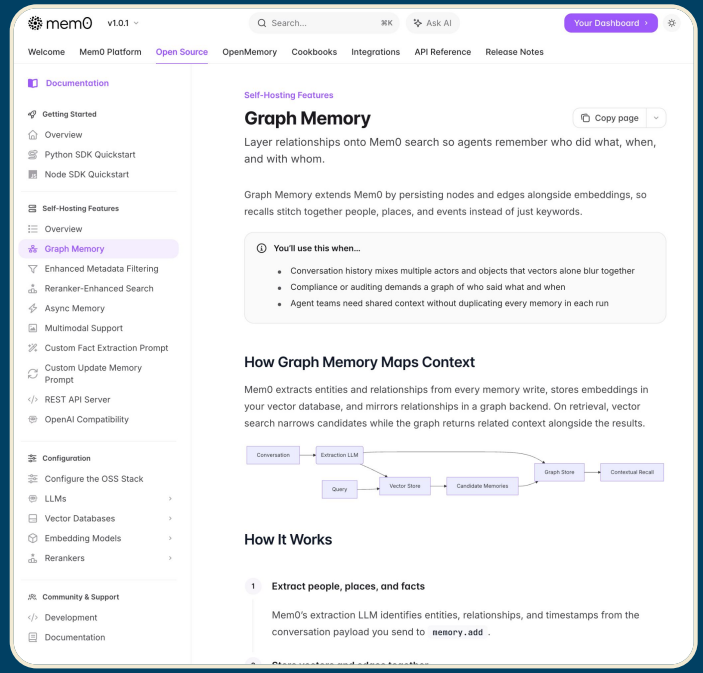
S3 - Episode 5

Neo4j Agent Memory

AI Agent Memory Landscape







https://neo4j.com/blog/developer/graphiti-knowledge-graph-memory/
https://neo4j.com/videos/neo4j-live-cognitive-sciences-and-dynamic-graphrag-2/
https://docs.mem0.ai/open-source/features/graph-memory

Neo4j Agent Memory: Context Graphs for AI Agents

README Apache-2.0 license

Neo4j Agent Memory

A graph-native memory system for AI agents. Store conversations, build knowledge graphs, and let your agents learn from their own reasoning -- all backed by Neo4j.

Neo4j Labs | Status Experimental | Support Community | pypi package 0.0.2 | python 3.10 | 3.11 | 3.12 | 3.13 | License Apache 2.0

> ⚠️ **This is a Neo4j Labs project.** It is actively maintained but not officially supported. There are no SLAs or guarantees around backwards compatibility and deprecation. For questions and support, please use the Neo4j Community Forum.

> **See it in action**: The Lenny's Podcast Memory Explorer demo loads 299 podcast episodes into a searchable knowledge graph with an AI chat agent, interactive graph visualization, geospatial map view, and Wikipedia-enriched entity cards.

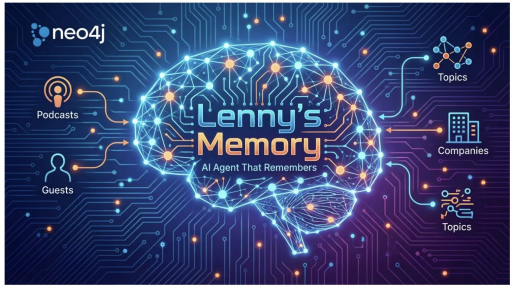
Features

- **Three Memory Types**: Short-Term (conversations), Long-Term (facts/preferences), and Reasoning (reasoning traces)
- **POLE+O Data Model**: Configurable entity schema based on Person, Object, Location, Event, Organization types with subtypes
- **Multi-Stage Entity Extraction**: Pipeline combining spaCy, GLiNER2, and LLM extractors with configurable merge strategies
- **Batch & Streaming Extraction**: Process multiple texts in parallel or stream results for long documents
- **Entity Resolution**: Multi-strategy deduplication (exact, fuzzy, semantic matching) with type-aware resolution
- **Entity Deduplication on Ingest**: Automatic duplicate detection with configurable auto-merge and flagging
- **Provenance Tracking**: Track where entities were extracted from and which extractor produced them
- **Background Entity Enrichment**: Automatically enrich entities with Wikipedia and Diffbot data

Meet Lenny's Memory: Building Context Graphs for AI Agents

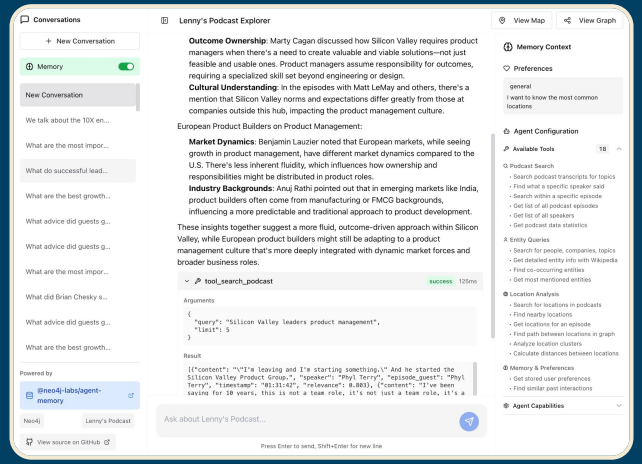
Meet Lenny's Memory: Complete Memory for AI Agents

William Lyon Follow 11 min read · Draft



lennys-memoryvercel.app

AI agents are transforming how organizations work — but they have a memory problem. Ask an agent about last week's conversation and you'll get



https://github.com/neo4j-labs/agent-memory
https://medium.com/neo4j/meet-lennys-memory-building-context-graphs-for-ai-agents-24cb102fb91a

Agent Memory API

Short-Term

Conversation history & session context

- add_message()
- search_messages()
- get_session()
- get_recent_messages()
- delete_session()

Long-Term

Knowledge graph of entities & relationships

- add_entity()
- search_entities()
- add_preference()
- search_locations_near()
- get_entity_graph()

Reasoning

Context graphs for explainability & learning

- start_trace()
- record_step()
- record_tool_call()
- get_similar_traces()
- get_trace_provenance()

Neo4j Context Graph

Vector Search + Graph Traversal

Entity Extraction Pipeline